

# A Multi-Vendor Corpus of Reference Low-Dose CT Reconstructions with Calibrated Per-Pixel Uncertainty and Signal-Equivalence Credentials

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## Abstract

Low-dose CT reconstruction research is rich in methods but poor in *shared, reusable derived data*: published methods release a model or a number, rarely the per-case reconstructed volumes, the per-pixel uncertainty that a clinical user would triage on, or a machine-checkable certificate of diagnostic equivalence. We describe a deposited corpus that supplies exactly this layer on top of the multi-vendor PWM-LDCT benchmark ([to be confirmed, 2027a](#)). For every test scan, at four dose levels (10%, 25%, 50%, 100%) and across multiple CT vendors, the corpus provides: (i) a **reference reconstruction** produced by an open, fully specified unrolled-iterative pipeline; (ii) a **per-pixel uncertainty map** from a five-model deep ensemble; (iii) the **paired per-pixel error map** against the full-dose reference, enabling calibration studies without re-running any model; (iv) **downstream-task output maps** (lung-nodule detection scores); (v) machine-readable **signal-equivalence credentials** computed under the framework of [to be confirmed \(2027b\)](#); and (vi) the matched reconstructions of five baseline methods for comparison. Every record is content-addressed, released under CC BY 4.0, and accompanied by an Apache-2.0 generating pipeline packaged as a pinned Docker RunBundle so the corpus is regenerable bit-for-bit within floating-point tolerance. Technical validation reports reconstruction fidelity, cross-vendor consistency, and—centrally for reuse—the agreement between the released uncertainty maps and the released error maps (Spearman  $\rho$ ). The corpus is intended for reuse in uncertainty-calibration research, cross-vendor robustness studies, downstream-task evaluation under dose reduction, and as a stable comparison baseline for new reconstruction methods.

## 1 Background & Summary

Deep-learning low-dose CT reconstruction has produced a decade of method papers reporting fractional-dB gains ([Chen et al., 2017](#); [Wang et al., 2020](#); [to be confirmed at submission, 2024](#)), yet the field shares almost no *derived* data. Two structural gaps follow. First, reconstructed volumes are seldom released: a researcher who wants to study how reconstruction error distributes across anatomy, dose, or vendor must retrain each method from scratch. Second, the two quantities most relevant to clinical deployment—*per-pixel uncertainty* and a *certificate of task-level equivalence*—are almost never published as data a third party can inspect. A deployer choosing a reconstruction for a new scanner, or a calibration researcher studying whether predicted variance tracks true error, has nothing reusable to build on.

The raw substrate for closing these gaps now exists: the PWM-LDCT benchmark ([to be confirmed, 2027a](#)) provides multi-vendor paired (full-dose, low-dose) CT, extending the field’s two single-vendor real-paired resources, AAPM-2016 ([AAPM, 2016](#)) and Mayo LDCT-PD ([McCollough et al., 2017](#)), both Siemens-only. What has been missing is the *derived layer*: a uniformly produced set of reconstructions, uncertainty maps, error maps, task outputs, and equivalence credentials over that substrate.

This Data Descriptor releases that layer. We generate every record with a single, fully specified open pipeline—an unrolled iterative reconstruction with a weight-shared denoiser and a five-model deep ensemble for uncertainty (Methods)—and deposit the outputs as a content-addressed corpus. We emphasise that the pipeline is a *means of generating reproducible data*, not the object of this paper; its architecture is reported in full only so the data are reproducible and the uncertainty maps are interpretable.

**Relationship to the PWM-LDCT benchmark (WS-1).** This corpus is deliberately disjoint from the underlying benchmark dataset. WS-1 releases the *raw paired CT scans*; the present corpus releases the *reconstruction, uncertainty, error, task-output, and credential records derived from them*. The two are intended to be cited together: WS-1 as the input data, this corpus as the reference derived data.

**Reuse value.** The corpus enables, without any model retraining: (1) **uncertainty-calibration research**—the paired (uncertainty map, error map) records are a ready-made benchmark for calibration metrics and recalibration methods; (2) **cross-vendor robustness studies**—reconstructions and errors are vendor-labelled, supporting leave-one-vendor-out analyses; (3) **downstream-task evaluation under dose reduction**—released task-score maps let others study how dose affects detection without re-deriving reconstructions; (4) **credential-based comparison**—the machine-readable signal-equivalence credentials are a stable target that new methods can report against under a content-addressed framework hash; and (5) a **stable comparison baseline**—five matched baseline reconstructions accompany the reference so new methods compare against identical inputs.

**Summary of what is released.** Table 1 summarises the record types; section 3 gives the full file-level manifest. All records, for all scans, at all dose levels and vendors, are deposited together under a single repository DOI ([[TODO: figshare/Zenodo DOI on deposit](#)]).

## 2 Methods

This section specifies how each released record was generated. It is included to make the corpus reproducible and the uncertainty maps interpretable; it is not a methodological contribution of this Data Descriptor.

### 2.1 Input data and provenance

Inputs are the PWM-LDCT ([to be confirmed, 2027a](#)) scans: paired full-dose and simulated/real low-dose acquisitions at signal ratios  $r \in \{0.10, 0.25, 0.50\}$  (with  $r = 1.00$  the full-dose reference), patient-level split into train/validation/test, vendor-labelled. Reference (target) images are the patient’s full-dose reconstruction. No scan crosses the train/test boundary at the patient level; all released records are for the held-out test split.

### 2.2 Reconstruction pipeline

Reconstructions are produced by an unrolled iterative scheme ([Adler and Öktem, 2018; Mardani et al., 2018](#)). Let  $\mathcal{R} : \mathbb{R}^{H \times W} \rightarrow \mathbb{R}^{n_{\text{views}} \times n_{\text{dets}}}$  be the Radon forward operator and  $\mathcal{R}^\dagger$  its filtered back-projection adjoint, both taken from the open `pwm_core.contrib.modalities.ct_radon` implementation ([PWM Protocol, 2026](#)) validated against the PWM modality test suite; we do not re-implement them, so any user regenerating the corpus inherits the identical, version-pinned forward model.

Given a low-dose sinogram  $y$ , the reconstruction  $\hat{x}$  is computed by  $K = 10$  data-consistency / denoising iterations (Algorithm 1). The denoiser  $f_\theta$  is a U-Net ( $\approx 4$  M parameters, four encoder / four decoder stages with (32, 64, 128, 256) channels, GELU, GroupNorm) whose weights are shared across iterations; the step size  $\tau$  is a learned scalar initialised at  $0.5/\|\mathcal{R}\|_2^2$ .

### 2.3 Uncertainty quantification

Each released reconstruction is the mean of a five-model deep ensemble ([Lakshminarayanan et al., 2017](#)): five instances of the architecture above, identical except for the random seed governing weight initialisation and

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**Algorithm 1** Unrolled iterative reconstruction (per ensemble member)

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```
1:  $x^{(0)} \leftarrow \text{TV-FBP}(y)$  ▷ TV-regularized warm start
2: for  $k = 0, 1, \dots, K-1$  do
3:    $x^{(k+1/2)} \leftarrow x^{(k)} - \tau \cdot \mathcal{R}^\dagger(\mathcal{R}(x^{(k)}) - y)$  ▷ Data consistency
4:    $x^{(k+1)} \leftarrow f_\theta(x^{(k+1/2)})$  ▷ Learned denoising
5: end for
6: return  $\hat{x} \leftarrow x^{(K)}$ 
```

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data ordering (seeds  $\{42, 43, 44, 45, 46\}$ ). For a low-dose input  $y$ , member  $m$  produces  $\hat{x}_m(y)$ ; the released point estimate and per-pixel uncertainty are

$$\hat{x}(y) = \frac{1}{5} \sum_{m=1}^5 \hat{x}_m(y), \quad \sigma(y)_{ij} = \sqrt{\frac{1}{5} \sum_{m=1}^5 (\hat{x}_m(y)_{ij} - \hat{x}(y)_{ij})^2}. \quad (1)$$

Deep ensembles are chosen over MC dropout because they calibrate better under distribution shift (Ovadia et al., 2019; Gal and Ghahramani, 2016) and have bounded ( $5\times$ ) inference cost. The released  $\sigma(y)$  map is the quantity intended for case-level triage; its *validity* as a triage signal is established in Technical Validation by comparison with the released error map.

## 2.4 Derived per-record quantities

For each test scan and dose level the pipeline emits, in addition to  $\hat{x}(y)$  and  $\sigma(y)$ : the **per-pixel error map**  $e(y)_{ij} = |\hat{x}(y)_{ij} - x_{ij}^{\text{ref}}|$  against the full-dose reference; a **lung-nodule detection score map** from a frozen pre-trained detector applied to  $\hat{x}(y)$ ; and a **signal-equivalence credential** computed with the `pwm_dose_equivalence` library (to be confirmed, 2027b) (5-tuple  $(r, T, \varepsilon, \alpha, \Pi)$  with verdict and confidence interval). The five baseline methods (Section 3) are run through the identical emit path so their records are field-for-field comparable.

## 2.5 Training and generation configuration

Models are trained on the PWM-LDCT training split with paired targets at  $r \in \{0.10, 0.25, 0.50\}$  mixed per minibatch with weights (0.4, 0.4, 0.2); augmentation is physical-consistency-preserving only (random flip  $p=0.5$ ; rotation  $[-5^\circ, 5^\circ]$ ; window-level shift  $[-50, 50]$  HU). Optimiser AdamW (Loshchilov and Hutter, 2019), cosine LR with 5% warmup, batch size 4 at  $512 \times 512$ , single A100; CuDNN deterministic, all RNGs seed-propagated. The complete hyperparameter table, library version pins, and compute disclosure accompany the deposit (Supplementary Table S1) so the corpus is regenerable.

## 3 Data Records

All records are deposited at [\[TODO: figshare/Zenodo DOI on deposit\]](#) under CC BY 4.0, mirrored to a content-addressed PWM RunBundle on IPFS (CID in Section 6) for integrity verification. The deposit is a single archive with the directory layout below; every leaf file is additionally listed with its SHA-256 in `MANIFEST.sha256` at the archive root.

### Directory layout.

```
pwm-ldct-recon-corpus/
  MANIFEST.sha256          # SHA-256 of every leaf file
  dataset_metadata.json    # ISA-style top-level metadata, DOI, licence
  reconstructions/
    <vendor>/<scan_id>/r<dd>/      # dd in {010,025,050,100}
      recon_mean.nii.gz          # reference reconstruction (this method)
```

| Record type              | Format             | Per-record content                                                                             |
|--------------------------|--------------------|------------------------------------------------------------------------------------------------|
| Reference reconstruction | NIFTI<br>(.nii.gz) | Ensemble-mean reconstruction $\hat{x}(y)$ , HU, $512 \times 512 \times Z$                      |
| Uncertainty map          | NIFTI              | Per-pixel ensemble std. dev. $\sigma(y)$ , same grid                                           |
| Error map                | NIFTI              | Per-pixel $ \hat{x} - x^{\text{ref}} $ vs full dose, same grid                                 |
| Task-score map           | NIFTI              | Frozen lung-nodule detector score map on $\hat{x}(y)$                                          |
| Credential               | JSON               | 5-tuple signal-equivalence credential (to be confirmed, 2027b) + verdict + CI + framework hash |
| Baseline reconstructions | NIFTI              | Same fields, for 5 baseline methods (Sec. 3.1)                                                 |
| Per-scan metadata        | JSON               | Vendor, dose ratio $r$ , anatomy, split, source-scan ID (links to WS-1)                        |

Table 1: Overview of released record types. Counts and total size per type are in Table 2; the full directory manifest is below.

```

uncertainty_sigma.nii.gz      # per-pixel ensemble std dev
error_abs.nii.gz              # |recon - full_dose_ref|
task_nodule_score.nii.gz      # frozen-detector score map
credential.json                # signal-equivalence credential
scan_meta.json                # vendor, r, anatomy, split, WS-1 source id
baselines/
  <method>/<vendor>/<scan_id>/r<dd>/ # method in {fbp,redcnn,transformer,
    recon.nii.gz                    #                diffusion,wgan_vgg}
    error_abs.nii.gz
    task_nodule_score.nii.gz
    credential.json
credentials/
  all_credentials.jsonl          # flat index of every credential.json
checksums_and_versions.json     # library version pins for regeneration

```

**Naming and indexing.** `scan_id` is the WS-1 source identifier, so any record back-links to its raw input. `r<dd>` encodes the dose ratio. The flat `all_credentials.jsonl` index lets a reuser query credentials (by vendor, dose, task, verdict) without walking the tree.

| Record type               | Count            | Format | Total size        |
|---------------------------|------------------|--------|-------------------|
| Reference reconstructions | [TODO: N]        | NIFTI  | [TODO: GB]        |
| Uncertainty maps          | [TODO: N]        | NIFTI  | [TODO: GB]        |
| Error maps                | [TODO: N]        | NIFTI  | [TODO: GB]        |
| Task-score maps           | [TODO: N]        | NIFTI  | [TODO: GB]        |
| Credentials               | [TODO: N]        | JSON   | [TODO: MB]        |
| Baseline reconstructions  | [TODO: N]        | NIFTI  | [TODO: GB]        |
| <b>Total</b>              | <b>[TODO: N]</b> | —      | <b>[TODO: GB]</b> |

Table 2: Record counts and sizes. Populated when the corpus is generated and deposited (WS-3 Phase 3, D9+270).

### 3.1 Baseline reconstructions included for comparison

To make the corpus a self-contained comparison resource, the matched reconstructions of five baselines are included, each run through the identical emit path: FBP; RED-CNN (Chen et al., 2017); WGAN-VGG (Yang et al., 2018); a transformer reconstruction (Wang et al., 2023); and a diffusion / score-based reconstruction (to be confirmed at submission, 2024). Each baseline is reproduced within  $\leq 0.5$  dB PSNR of its published numbers before its records are emitted; reproduction recipes accompany the deposit.

## 4 Technical Validation

Validation establishes that the released records are fit for reuse. It is organised around the three properties a reuser depends on: that the reconstructions are faithful, that they are consistent across vendors, and that the released uncertainty maps actually track the released error maps.

**Reconstruction fidelity.** Table 3 reports PSNR, SSIM, and LPIPS (Zhang et al., 2018) of every released reconstruction against its full-dose reference, per dose level, for the reference method and the five baselines. These quantify how close each released volume is to ground truth; they are descriptive data-quality evidence, not a competition.

| Method           | $r = 0.10$ |          | $r = 0.25$ |          | $r = 0.50$ |          |
|------------------|------------|----------|------------|----------|------------|----------|
|                  | PSNR       | SSIM     | PSNR       | SSIM     | PSNR       | SSIM     |
| FBP              | [TODO: ]   | [TODO: ] | [TODO: ]   | [TODO: ] | [TODO: ]   | [TODO: ] |
| RED-CNN          | [TODO: ]   | [TODO: ] | [TODO: ]   | [TODO: ] | [TODO: ]   | [TODO: ] |
| WGAN-VGG         | [TODO: ]   | [TODO: ] | [TODO: ]   | [TODO: ] | [TODO: ]   | [TODO: ] |
| Transformer      | [TODO: ]   | [TODO: ] | [TODO: ]   | [TODO: ] | [TODO: ]   | [TODO: ] |
| Diffusion        | [TODO: ]   | [TODO: ] | [TODO: ]   | [TODO: ] | [TODO: ]   | [TODO: ] |
| <b>Reference</b> | [TODO: ]   | [TODO: ] | [TODO: ]   | [TODO: ] | [TODO: ]   | [TODO: ] |

Table 3: Reconstruction fidelity of the released records against the full-dose reference, with 95% bootstrap CIs ( $B=1000$ ) over test patients. Populated at deposit (D9+270).

**Cross-vendor consistency.** Because records are vendor-labelled, we validate that fidelity does not collapse across vendors. Table 4 reports the PSNR difference  $\Delta$  under leave-one-vendor-out generation (the held-out vendor’s scans reconstructed by models trained without that vendor). This characterises the regime in which the released cross-vendor records are trustworthy for reuse versus extrapolative.

[TODO: TABLE: Cross-vendor PSNR difference ( $\Delta$ , dB) under leave-one-vendor-out generation; reported so reusers know which vendor records are interpolation vs extrapolation]

Table 4: Cross-vendor PSNR difference ( $\Delta$ , dB) under leave-one-vendor-out generation; reported so reusers know which vendor records are interpolation vs extrapolation (placeholder; populate when the corpus is deposited).

**Uncertainty–error agreement (central reuse validation).** The corpus’s distinctive reuse value is the paired (uncertainty, error) records. We validate that the released  $\sigma(y)$  tracks the released error  $e(y)$  by the Spearman correlation between them, per pixel and aggregated per case, on the held-out test split (Figure 1, Table 5). A high correlation certifies the uncertainty maps as a usable triage signal; a low correlation is itself reported so reusers do not over-trust the variance. This is the single most important validation for the corpus’s stated calibration-research reuse.

| Dose       | Spearman $\rho$ (per-pixel) | Spearman $\rho$ (per-case mean) | 95% CI   |
|------------|-----------------------------|---------------------------------|----------|
| $r = 0.10$ | [TODO: ]                    | [TODO: ]                        | [TODO: ] |
| $r = 0.25$ | [TODO: ]                    | [TODO: ]                        | [TODO: ] |
| $r = 0.50$ | [TODO: ]                    | [TODO: ]                        | [TODO: ] |

Table 5: Agreement between released uncertainty maps and released error maps. Populated at deposit (D9+270).

[TODO: FIGURE: Reliability diagram: released per-pixel ensemble standard deviation vs observed per-pixel reconstruction error on the held-out test split.]

Figure 1: Reliability diagram: released per-pixel ensemble standard deviation vs observed per-pixel reconstruction error on the held-out test split. (placeholder; populate when the corpus is deposited).

**Downstream-task validity.** Table 6 reports lung-nodule detection AUC (fixed FPR = 0.1) computed from the released task-score maps with a frozen detector held constant across all methods. This validates that the released reconstructions preserve diagnostically relevant structure, and documents the known divergence between pixel fidelity and task performance (Kim et al., 2019) so reusers interpret both layers correctly.

[TODO: TABLE: Lung-nodule detection AUC (frozen detector, fixed FPR = 0.1) computed from the released task-score maps, per method, per dose level]

Table 6: Lung-nodule detection AUC (frozen detector, fixed FPR = 0.1) computed from the released task-score maps, per method, per dose level (placeholder; populate when the corpus is deposited).

**Credential validation.** Each released credential is internally audited with the `pwm-audit` tool (to be confirmed, 2027b) (schema validity, verdict-vs-CI self-consistency, framework-hash recognition); Table 7 summarises the verdicts. The audit is bundled in the deposit so a reuser can re-verify every credential without re-running the bootstrap.

| $r$  | Task              | $\epsilon$ | $\alpha$ | Subpopulation            | Verdict  | CI       |
|------|-------------------|------------|----------|--------------------------|----------|----------|
| 0.25 | Lung nodule AUC   | 0.05       | 0.05     | adult-chest test split   | [TODO: ] | [TODO: ] |
| 0.10 | Lung nodule AUC   | 0.05       | 0.05     | adult-chest test split   | [TODO: ] | [TODO: ] |
| 0.25 | Liver lesion Dice | 0.03       | 0.05     | adult-abdomen test split | [TODO: ] | [TODO: ] |

Table 7: Signal-equivalence credentials (to be confirmed, 2027b) for the reference reconstruction against full-dose, audited with `pwm-audit`. Verdicts: `PASS` / `FAIL` / `INDETERMINATE`. Note: the WS-1 test cohort sits in the `INDETERMINATE`-dominated regime for the AUC default; absence of `PASS` reflects cohort size, not non-equivalence (see to be confirmed, 2027b).

**Reproducibility of generation.** The full corpus regenerates from the deposited Apache-2.0 RunBundle: `docker run` on a fresh machine reproduces every record bit-identically within IEEE-754 tolerance (seeds, library pins, and CuDNN-deterministic settings in Supplementary Table S1). The RunBundle is independently re-executed by the PWM challenge scoring service before posting, providing third-party regeneration evidence.

## 5 Usage Notes

**Loading.** All image records are NIfTI and load with any standard reader (`nibabel`, `ITK`); credentials are JSON validated against the published schema (to be confirmed, 2027b). The `all_credentials.jsonl` index supports tabular queries by vendor, dose, task, and verdict. Each record’s `scan_meta.json` links back to its WS-1 raw source (to be confirmed, 2027a).

**Intended reuses.** (1) Uncertainty-calibration research: treat the paired (`uncertainty_sigma`, `error_abs`) maps as a ready calibration benchmark. (2) Cross-vendor robustness: use vendor labels and the leave-one-vendor-out validation to scope claims. (3) Dose-dependent downstream-task studies: use the released task-score maps without re-deriving reconstructions. (4) Method comparison: benchmark a new reconstruction against the matched baseline records on identical inputs and report a credential against the same framework hash.

**Cautions for reuse.** The released uncertainty is ensemble disagreement; it can be non-informative where all members fail in correlated ways (out-of-distribution anatomy)—inspect the reliability diagram before triaging on  $\sigma$ . Cross-vendor records for vendors not in training are extrapolation, not measurement. Dose ratios below 0.10 are not characterised. Reconstructions are single-slice 2D. Credential INDETERMINATE verdicts on this cohort reflect sample size (to be confirmed, 2027b), not evidence of non-equivalence, and should not be read as a negative finding about a method.

**Licensing.** Data records: CC BY 4.0. Generating code and RunBundle: Apache 2.0. Reuse requires citing this Data Descriptor and the WS-1 source dataset (to be confirmed, 2027a).

## 6 Code Availability

The complete generating pipeline—training, inference, ensemble aggregation, error/task/credential emission, and the evaluation harness that produced every Technical Validation number—is released under Apache 2.0 at [https://github.com/integritynoble/low\\_dose\\_CT/tree/main/WS-3\\_reference\\_method](https://github.com/integritynoble/low_dose_CT/tree/main/WS-3_reference_method), and packaged as a pinned Docker RunBundle (CID [TODO: IPFS CID at deposit]; anchored by a PWM L4 cert for integrity). Pre-trained ensemble weights, the pinned `Dockerfile`, evaluation scripts, random seeds, and the training-compute disclosure (Supplementary Table S1) permit bit-identical regeneration of the corpus within floating-point tolerance on equivalent hardware. The forward model is the upstream `pwm_core.contrib.modalities.ct_radon` (PWM Protocol, 2026); the credentialing library is `pwm_dose_equivalence` (to be confirmed, 2027b).

## Data Citations

The deposited corpus: [TODO: figshare/Zenodo DOI on deposit] (CC BY 4.0).

Source benchmark dataset (WS-1): to be confirmed (2027a).

Signal-equivalence framework and credential schema: to be confirmed (2027b).

## Acknowledgments

[TODO: Fill at submission.]

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